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Economic impacts of near-optimal solutions with non-convex pricing

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ABSTRACT

We explore the relationship between consumer and producer surplus and optimality gaps in mixed-integer linear programming applied to electricity markets. We analyze a long-run adapted resource mix and introduce flexible demand. This allows for comparison of total producer and consumer surplus achieved across near-optimal solutions under different non-convex pricing models. Results indicate that for pricing models dependent on the primal solution, consumer surplus generally increases (although not monotonically) with increasing optimality and producer surplus correspondingly decreases. Short-run cost recovery is also possible over the operating horizon without make-whole payments for all models examined. These results are highly influenced by scarcity rent from price-setting elastic demand.

1. Introduction

Determining optimal prices in markets with non-convex costs in which optimality gaps in the dispatch decision remain poses a continuing challenge. Since the beginning of the transition to liberalized wholesale electricity markets from vertically integrated utilities, it has been known that optimality gaps that negligibly affect the objective value may nevertheless significantly impact the profitability of individual resources [1]. As Lagrangian relaxation algorithms gave way to tractable algorithms to solve a mixed-integer programming (MIP) formulation, optimality gaps decreased. The move from Lagrangian relaxation approaches to MIP solvers was motivated by the goal of finding better solutions and thus improving social welfare, or social surplus [2]. The question that naturally follows is, how important is finding a solution closer to optimality for MIP solvers in terms of social surplus? System operators typically solve day-ahead markets with unit commitment (UC) within some optimality tolerance, e.g., a 0.1% relative MIP gap in MISO [2].

The volume of surplus transfers between producers is not monotone in the size of the optimality gap, i.e., it does not strictly decrease as the solution approaches optimality [3]. Authors in [3] find in a test system over a 24-h period with locational marginal pricing that make-whole payments, which ensure units recover all short-run costs, reduce the volatility of intra-producer transfers. It is then explored in [4] whether findings in [3] are still supported over a longer time frame. A daily rolling-horizon with look-ahead UC model is solved in a large test system, obtaining near-optimal solutions at several different MIP gap targets. Producer surplus of individual generators is shown to still vary considerably even with solutions closer to optimality. When including elastic demand, solutions closer to optimality are shown to be closer to the demand cleared in the MIP optimal solution. Authors in [5] examine redistribution of total surplus with inelastic demand in different near-optimal solutions, demonstrating that pricing models that do not depend on the primal UC solution, e.g., convex hull pricing, are associated with smaller wealth transfers.

It is important to distinguish between changes in intra-producer surplus transfers and changes in total producer and consumer surplus. Producer surplus is simply profit, and intra-producer surplus transfer implies that certain producers benefit at the expense of other producers. Total producer surplus, on the other hand, represents the total surplus achieved by producers under a given UC dispatch with a given pricing model. Taking the perspective of a central operator, another concern is the total amount of consumer surplus (the difference between the benefit of cleared demand and payments to producers). Previously in [6], we target two different MIP gap tolerances and find that resource mixes adapted in the long-run to different non-convex pricing models can vary based on which target was used. While not explored explicitly, results in [7] similarly suggest an increase in overall consumer surplus as optimality improves.

Instead of primarily focusing on transfers between producers in the short-run as in [3–5], we turn our attention to the effect of near optimal solutions on transfers between total consumer and producer surplus in the long-run under different non-convex pricing models. We explore a set of near-optimal solutions across a range of MIP gaps and examine the relationships between consumer and producer surplus in a system with partially elastic demand. We consider only systems in a long-run equilibrium resource mix. This is because in non-adapted mixes,

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the presence of short-run losses (that are typically compensated by make-whole payments) may be an appropriate signal to exit the market as opposed to an issue arising from non-convex pricing. Analyzing long-run equilibria also allows us to examine not simply short-run profits, but long-run cost recovery of producers. Fixing the underlying resource mix, we then examine near-optimal solutions of the short-run UC dispatch and apply several pricing models. We include scenarios that require daily short-run cost recovery as well as scenarios only enforcing short-run cost recovery over the lifetime of the generator.

2. Long-run optimal resource mix

From classical marginal pricing theory, we know that an idealized central planner (CP) maximizing social surplus would choose the same resource mix as a perfectly competitive market with system-wide marginal pricing.[1] At this equilibrium, no units make a profit or loss and all social surplus is consumer surplus. Marginal prices do not exist in markets with non-convex costs, but we can still find a benchmark CP resource mix that maximizes benefit to consumers and minimizes producer costs with a capacity expansion model. We can decrease the impact of lumpy investments by using a large test system. Our aim is to solve to a sufficiently small optimality gap such that we find a stable resource mix, acknowledging that because of lumpy investments, we may have more than one long-run equilibrium. In a market with non-convex costs, there may be incentives via profits and losses to move away from this benchmark CP resource mix to a mix that is adapted to a given pricing model. However, because we would like to compare different pricing models across the same set of near-optimal solutions, we keep the same underlying CP resource mix. Below we define the notation and model formulation and refer to [6] for a more in-depth description.

2.1. Nomenclature

Indices and Sets

$g \in G$	Set of generators
$G^T \subseteq G$	Set of thermal generators
$G^N \subseteq G^T$	Set of nuclear generators
$G^V \subseteq G$	Set of VRE resources
$t \in T$	Set of time periods (h)
$l \in L$	Set of demand bids

Parameters

C_g	Variable cost (\$/MWh)
F_g	Startup cost (\$)
C_g^{inv}	Annualized investment cost of generator g (\$/yr)
P_g^{min}	Minimum operating capacity (MW)
P_g^{max}	Maximum operating capacity (MW)
M_g^{on}	Minimum on time (h)
M_g^{off}	Minimum off time (h)
R_g	Maximum ramp up/down rate (MW/h)
$\mathcal{P}_{tg}$	Maximum output for VRE resource (MW)
B_l	Value of demand bid l (\$/MWh)
D_{tl}	Maximum quantity of demand bid l at time t (MW)

Variables

p_{tg}	Committed generation for generator g at time t (MW)
u_{tg}	(Binary) commitment status for generator g at time t
z_{tg}	(Binary) startup decision for generator g at time t
y_{tg}	(Binary) shutdown decision for generator g at time t
d_{tl}	Amount of cleared demand bid l at time t (MW)
x_g	Binary build decision for each generator $g \in G^T$

[1] See [8] for proof of equivalent optimality conditions in the context of electricity markets.

2.2. Capacity expansion model

The factor Θ scales up short-run costs to an annualized level in objective function (1a) that maximizes benefits and minimizes costs, assuming no transmission congestion. Constraint (1b) ensures power balance with cleared demand bids. The remaining constraints define operating characteristics of generators, exogenous variable renewable energy (VRE), and reflect an expectation of one startup cost for nuclear in a year included in investment cost.

$$\max_{(x,p,u,z,y,d)} \quad \Theta \sum_{t\in T}\sum_{l\in L} B_l d_{tl} - \sum_{g\in G} C_g^{inv} x_g \\ -\Theta \sum_{t\in T}\sum_{g\in G} (C_g p_{tg} + F_g z_{tg}) \tag{1a}$$

s.t.

$$\sum_{g\in G} p_{tg} = \sum_{l\in L} d_{tl} \qquad \forall t \in T \tag{1b}$$

$$0 \le d_{tl} \le D_{tl} \qquad \forall t \in T, l \in L \tag{1c}$$

$$p_{tg} \le P_g^{max} x_g \qquad \forall t \in T, g \in G^T \tag{1d}$$

$$z_{tg} + y_{tg} \le 1 \qquad \forall t \in T, g \in G^T \tag{1e}$$

$$u_{tg} - u_{t-1,g} = z_{tg} - y_{tg} \qquad \forall t \in 2...T, g \in G^T \tag{1f}$$

$$z_{tg} = u_{tg} \qquad \forall t = 1, g \in G^T : g \notin G^N \tag{1g}$$

$$z_{tg} = 0 \qquad \forall t = 1, g \in G^N \tag{1h}$$

$$y_{tg} = 0 \qquad \forall t = 1, g \in G^T \tag{1i}$$

$$z_{tg} + \sum_{t'=t+1}^{\min(t+M_g^{on}-1,T)} y_{t'g} \le 1 \\ \forall t \in 1...T-1, g \in G^T : M_g^{on} > 1 \tag{1j}$$

$$y_{tg} + \sum_{t'=t+1}^{\min(t+M_g^{off}-1,T)} z_{t'g} \le 1 \\ \forall t \in 1...T-1, g \in G^T : M_g^{off} > 1 \tag{1k}$$

$$P_g^{min} u_{tg} \le p_{tg} \le P_g^{max} u_{tg} \qquad \forall t \in T, g \in G^T \tag{1l}$$

$$-R_i \le p_{tg} - p_{t-1,g} \le R_i \qquad \forall t \in T, g \in G^T \tag{1m}$$

$$0 \le p_{tg} \le \mathcal{P}_{tg} \qquad \forall t \in T, g \in G^V \tag{1n}$$

$$p_{tg} \ge 0 \qquad \forall t \in T, g \in G \tag{1o}$$

$$u_{tg}, z_{tg}, y_{tg} \in \{0,1\} \qquad \forall t \in T, g \in G^T \tag{1p}$$

$$x_g \in \{0,1\} \qquad \forall g \in G^T \tag{1q}$$

3. Short-run dispatch and pricing models

3.1. Short-run dispatch model

Once a capacity mix is fixed, the short-run UC dispatch is solved. This formulation replaces the objective function (1a) with (2):

$$\max_{(p,u,z,y,d)} \quad \sum_{t\in T}\sum_{l\in L} B_l d_{tl} - \sum_{t\in T}\sum_{g\in G} (C_g p_{tg} + F_g z_{tg}) \tag{2}$$

The subset of units G is redefined as only units that were built in the capacity expansion model, $g \in G : x_g = 1$. Constraints (1d) and (1q) are omitted.

3.2. Pricing models

Pricing models are formed from adjustments to the short-run dispatch model after a solution has been found. Prices are found as the dual variables of the power balance constraints from the pricing model runs.

- **Fixed configuration pricing (FCP)**: The binary variables are relaxed and fixed to the values determined in the short-run dispatch model [9]:

$$\{u, z, y\} \in [0, 1] \tag{3}$$

$$u, z, y = u^*, z^*, y^* \tag{4}$$

This method is also called locational marginal pricing, although locational prices could be calculated for any of the models and true marginal prices do not exist in the non-convex setting.

- **Convex hull pricing (CHP)**: This method seeks to find the marginal prices corresponding to the convex hull of the non-convex optimal value function, minimizing lost opportunity costs [10]. An approximation that is exact with a tight UC formulation in the case of no binding ramping constraints (see [11]) is to simply relax the binary variables of the short-run dispatch model:

$$\{u, z, y\} \in [0, 1] \tag{5}$$

- **Relaxed P_g^{min} operation (RPM)**: The minimum operation level is relaxed such that $0 \le p_g \le P_g^{max} u_g \; \forall g \in G : u_g^* = 1$. Binary variables are relaxed and fixed to the previously-found optimal values:

$$\{u, z, y\} \in [0, 1] \tag{6}$$

$$u, z, y = u^*, z^*, y^* \tag{7}$$

A form of this model relaxing minimum operation for a subset of online units is used in NYISO and several other regions, motivated by block-loaded units being unable to set the marginal price under FCP [12].

4. Consumer and producer incentives and optimality gaps

4.1. Consumer and producer incentives

With market-clearing prices λ and side payments s, we define short-run profit Π, initial short-run profit Π^0, and short-run costs Ξ as follows:

$$\Pi_g := \sum_{t \in T} ((\lambda_t - C_g) p_{tg} - F_g z_{tg} + s_{tg}) \tag{8}$$

$$\Pi_g^0 := \Pi_g - \sum_{t \in T} s_{tg} \tag{9}$$

$$\Xi_g := \sum_{t \in T} (C_g p_{tg} + F_g z_{tg}) \tag{10}$$

We define long-run profits as a percentage of total costs:

$$\pi_g := \frac{\Theta \Pi_g - C_g^{inv}}{\Theta \Xi_g + C_g^{inv}} \tag{11}$$

Social surplus is the sum of producer and consumer surplus. Total thermal generator producer surplus is defined as

$$PS := \sum_{g \in G^T} (\Pi_g - \frac{1}{\Theta} C_g^{inv}) \tag{12}$$

Total consumer surplus is given by

$$CS := \sum_{t \in T} (\sum_{l \in L} B_l d_{tl} - \sum_{g \in G} \lambda_t p_{tg} - \sum_{g \in G^T} s_{tg}) \tag{13}$$

For a pricing model to support the central dispatch decision, the total compensation must be such that no perceived losses exist. Perceived losses are lost opportunity costs (LOC), the difference between what a unit could make given a price if able to schedule its own dispatch (its preferred profit) and what it would make with the same price following the centralized dispatch decision, plus any additional compensation received as side payments. We define initial lost opportunity costs LOC^0 without side payments and final LOC as

$$LOC_g^0 := \max_{u_g, p_g} \quad \Pi_g^0(\lambda^*, u_g, p_g) - \Pi_g^0(\lambda^*, u_g^*, p_g^*) \tag{14}$$

$$LOC_g := LOC_g^0 + \sum_{t \in T} s_{tg} \tag{15}$$

A subset of LOC^0 is make-whole payments (MWP), the revenue required for short-run revenue adequacy. This occurs when the unit would prefer to not operate at the market-clearing price. MWP are the most common form of side-payment in non-convex pricing and are defined over some time horizon as

$$MWP_g := -\min(0, \Pi_g^0(\lambda^*, u^*, p^*)) \tag{16}$$

Typically, MWP are provided as needed on a daily basis. However, there is no a priori reason why a system in a long-run adapted equilibrium should have no units making a loss on any given day. It may be most optimal, for instance, for a unit to suffer a loss one day but gain sufficient inframarginal rent over its operating life to have no long-run losses. Nevertheless, a daily MWP for a market that clears daily may be a necessary incentive for units to bid truthfully and follow market dispatch instructions. For these reasons we use both a typical daily MWP requirement as the default in all pricing models (MWP_D) and alternatively a guarantee of no operational losses over the entire operating horizon (MWP_H).

4.2. Relative MIP gap

A relative MIP gap is defined for a minimization problem as the difference between the lower and upper objective bound normalized by the incumbent objective value (the upper bound). If z_P is the primal objective bound and z_D the dual, then

$$\text{MIP gap} := \left| \frac{z_P - z_D}{z_P} \right| \tag{17}$$

Relative MIP gaps depend on the problem formulation. Because of this, we must be wary about comparing relative MIP gaps between optimal dispatch problems that do and do not explicitly include benefit of cleared demand in the objective value. For example, let C be a vector of costs associated with the vector of production decisions p and B the benefit of cleared demand d. We then have the objective function

$$\max_{p,d} \quad B^\top d - C^\top p$$

If we assume inelastic demand and that all demand is served, then $B^\top d$ cancels out in the numerator of the associated MIP gap but is still present in the denominator:

$$\left| \frac{C^\top p_D - C^\top p_P}{B^\top d_P - C^\top p_P} \right| \tag{18}$$

If $B^\top d$ is large relative to $C^\top p$, then an objective value considering the benefit to demand will result in a much smaller relative MIP gap than an objective value that excludes this term, even though the producer costs are equivalent. The ratio of total benefit to total cost gives some insight as to how much smaller a relative MIP gap to expect.

5. Results

5.1. Test case

Scenarios are constructed using correlated demand, wind generation, and solar generation profiles from [13]. We scale demand and VRE profiles by a factor of 10, yielding a system with peak load of approximately 47.5 GW over the 4 sample weeks selected. Non-coincident peak wind is over 50% of peak load, and non-coincident peak solar is over 25%. VRE is treated as exogenous, and we perform a greenfield capacity

Table 1
Technical parameters for thermal generators.

Tech	Min output (MW)	Max output (MW)	Ramp Up/Down (MW/h)	Up/Down time (h)
Nuclear	900	1000	190	36
CCGT	150	400	320	3
OCGT	50	200	360	0

Table 2
Cost parameters for thermal generators.

Tech	Investment (M$/GW-yr)	Startup (M$)	Variable ($/MWh)
Nuclear	489	1	6.5
CCGT	129	0.06	58.5
OCGT	106	0.01	99.4

expansion for thermal generators of three technology types: nuclear, CCGT, and OCGT, representing base, intermediate, and peaking units. Technical parameters are given in Table 1 and cost parameters are given in Table 2. Parameters are adapted from data used in [14]. We use partially price-responsive demand assuming 90% of demand at each time interval is inelastic (with a benefit of $10,000/MWh) and 10% of demand is elastic, represented by 200 equally-sized bids descending in price from $10,000/MWh to $50/MWh as in [15], which notes that recent PJM market results indicate such a level of responsive demand may be realistic [16].

All models are solved with Gurobi. The capacity expansion model yields a benchmark CP resource mix of 9 nuclear, 57 CCGT, and 67 OCGT units, for a total of 133 thermal units. The generator mix in the CP solution is determined by targeting a MIP Gap of 0 with a time limit, yielding a MIP gap of 0.0003%.[2] Next we use Gurobi's solution pool option to find a set of near-optimal solutions using the short-run dispatch model with a single horizon of 4 weeks, the same as the capacity expansion model.[3] The near-optimal solutions are required to be within 1% of the objective value of the best solution found within a target MIP gap of 0.02%. We find 51 qualifying near-optimal solutions.[4] The ratio of benefit of cleared demand to total producer costs at the best solution with FCP is approximately 200:1, which leads to very small relative MIP gaps. All pricing runs are convex, but calculating lost opportunity costs does require solving additional MIP problems. For these we maintain Gurobi's default MIP gap setting of 0.01%.[5]

5.2. Consumer surplus and producer surplus

We find that for the pricing models that depend on the primal solution, the total consumer surplus tends to increase with optimality while the total producer surplus tends to decrease, although not monotonically. Fig. 1 shows how consumer surplus changes with each near-optimal solution ordered from worst to best, with Solution 1 being the best solution found. Results are shown for the pricing models FCP, CHP, and RPM with the default MWP_D as well as the alternative MWP_H. While as expected the relationship is not monotonic, there is nevertheless a significant trend for FCP and RPM, which depend on the primal solution. RPM is found to result in the lowest consumer surplus

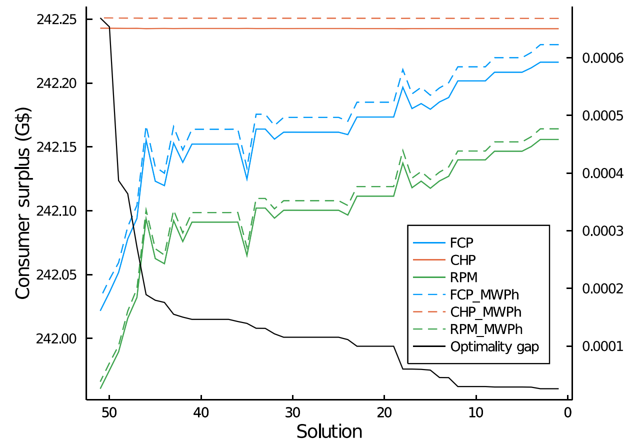

Fig. 1. Consumer surplus and optimality gap across near-optimal solutions by pricing model.

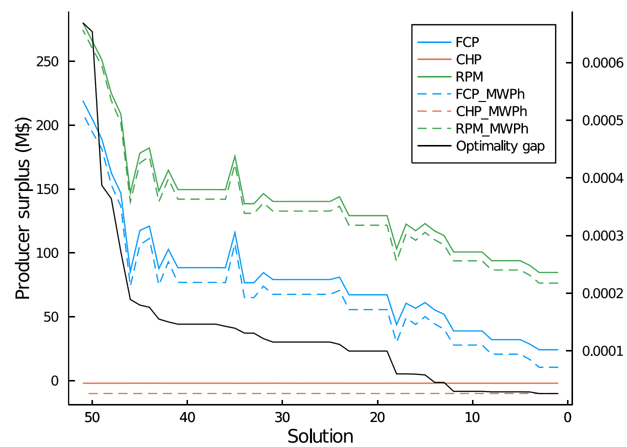

Fig. 2. Long-run producer surplus and optimality gap across near-optimal solutions by pricing model.

followed by FCP, with both substantially improving with optimality. CHP yields the highest consumer surplus. Fig. 2 shows that producer surplus correspondingly trends downward for primal-dependent models as the solution's optimality gap decreases. Note that producer surplus is reported only for thermal units. This trend does not change if we include MWP_D or if we do not enforce daily short-run cost recovery (MWP_H). Recall that in a convex long-run adapted system there would be no producer surplus. Between the least and most optimal solution found, producer surplus can decrease considerably, e.g., for FCP there is a reduction of 88% between the worst and best solution. Surplus under CHP does not change perceptibly across near-optimal solutions; while the amount of demand cleared and generators online may change between near-optimal solutions, the market-clearing prices found do not since CHP does not depend on the primal solution. We see through the illustration of a sample day in Section 5.4 that the different market-clearing prices are a primary driver of differences in surplus between near-optimal solutions for primal-dependent pricing models.

5.3. Producer profits and incentives

Short-run revenue adequacy is achieved in all pricing models tested without side payments if enforced over the entire solution horizon (MWP_H). If we instead require short-run revenue adequacy for each day (MWP_D), then side payments in the form of MWP are required. No substantial differences are seen in MWP for thermal units across near-optimal solutions, as seen in Fig. 3. When short-run cost recovery is

[2] Due to the binary build decisions of units of fixed size, it is possible that more than one long-run optimal resource mix exists.

[3] We assume perfect foresight so that uncertainty and rolling horizons do not additionally distort producer profits beyond the impact of non-convexities.

[4] Gurobi's pool search mode is set such that the solver seeks to find the N best solutions. We specify a high value of N, in attempt to find many qualifying solutions that differ in values for integer variables. However, a diversity criterion is not used.

[5] Note that the preferred profit function only includes producer profits and not benefit of cleared demand.

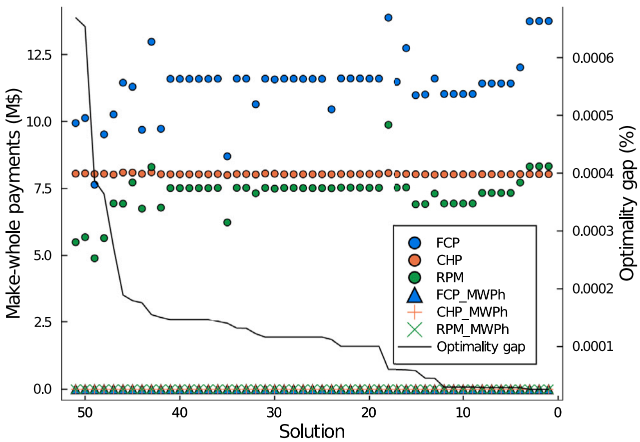

Fig. 3. Make-whole payments across near-optimal solutions.

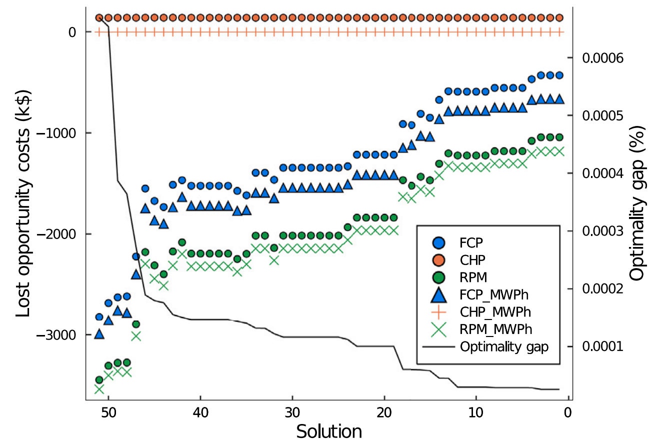

Fig. 4. Remaining lost opportunity costs after transfer of any side payments across near-optimal solutions.

required daily, MWP for RPM are typically lower and less volatile than MWP for FCP. MWP for CHP are consistent across solutions (note that CHP minimizes LOC, not MWP). For primal-dependent pricing models, we see a trend of less extreme LOC as the solution becomes more optimal, illustrated in Fig. 4. Note that Fig. 4 displays LOC remaining after any MWP have been transferred, and that perceived losses are shown as negative. CHP with MWP_H yields no LOC, as this method seeks to minimize this incentive to deviate from the schedule. However, if MWP_D are provided, producers actually perceive an excess profit. Producers are thus overcompensated relative to what is necessary for them to be fully incentivized to follow the central dispatch decision. Future work should explore if this issue persists when CHP is calculated on a rolling horizon basis.

By examining long-run profitability by technology type, we can see how near-optimal solutions impact intra-producer wealth transfers. Fig. 5 shows the average long-run profit of each technology type for the default pricing models (with MWP_D). Recall that in a convex market in long-run equilibrium, no producers would make a profit or loss. Here we see producers making both profits and losses due to lumpy investments, non-convex costs, and also the CP solution not necessarily being the long-run adapted mix for any particular non-convex pricing model. For primal-dependent models FCP and RPM, there is a notable downward trend of profits for all technology types as the solution's optimality improves. This mirrors the trend seen in total producer surplus in Fig. 2. In this system, the relative profits of technology types do not vary substantially with optimality, but rather uniformly decrease, indicating there is not significant transfer of producer surplus between technology types. CHP provides the closest profits to achieve long-run cost recovery without overcompensation, but the peaking OCGT technology type still suffers a small loss in the long-run. Comparing long-run profits between the MWP_D and MWP_H scenarios in Fig. 6, we see relatively small changes, as the MWP_D provide only a small amount of additional income.

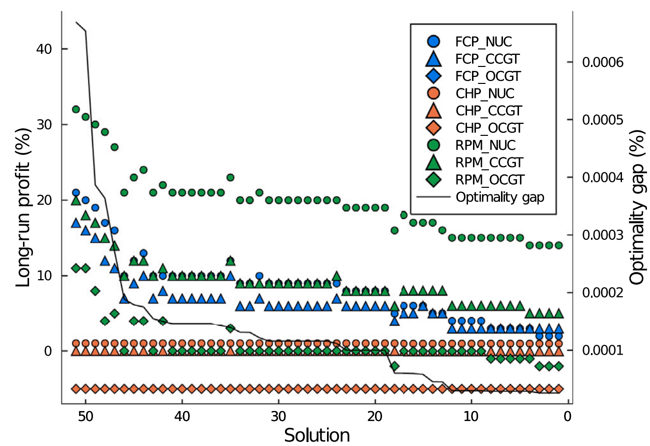

Fig. 5. Long-run profits as percent of overall costs by technology type with MWP_D.

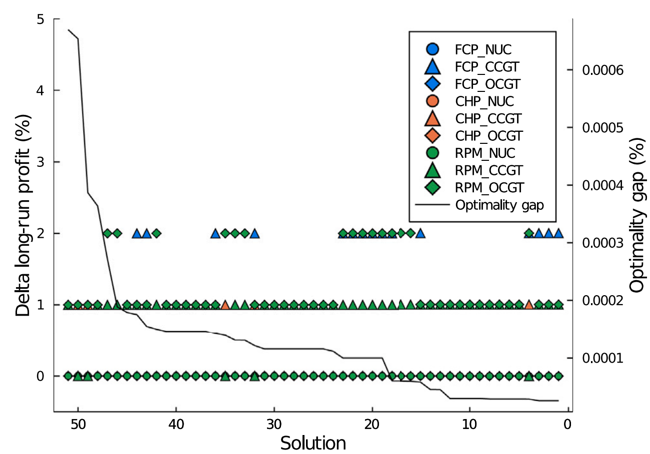

Fig. 6. Long-run profits percentage point delta between MWP_D and MWP_H cases.

5.4. Comparison of near-optimal solutions

To understand why near-optimal solutions generally result in superior outcomes in consumer and producer surplus with increasing optimality, it is helpful to examine an illustrative day. Fig. 7 shows the production schedule by technology type and the FCP market clearing price over a single day for the best solution found, Solution 1. Fig. 8 shows the same for a solution with a higher optimality gap, Solution 50. Fig. 9 shows the difference in unit commitment decisions and in cleared demand between these two solutions. The production schedules are largely similar. However, the two solutions make slightly different commitment decisions, leading to different levels of cleared demand and different prices in several evening hours, given in Table 3. In these hours, the price is being set by the elastic portion of demand in both solutions. Because Solution 1 clears a higher level of demand than Solution 50, the price is lower. This results in higher consumer surplus both from the higher benefit of cleared demand and the lower market clearing price. Conversely, producers receive less compensation than under Solution 50, which is less optimal and has a higher total

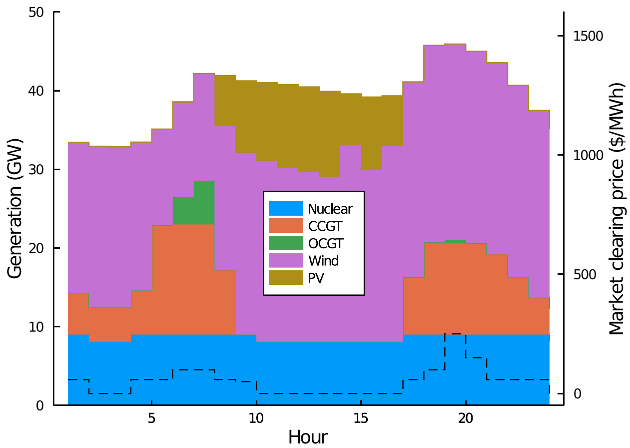

Fig. 7. Production by technology type and FCP market-clearing price for Solution 1.

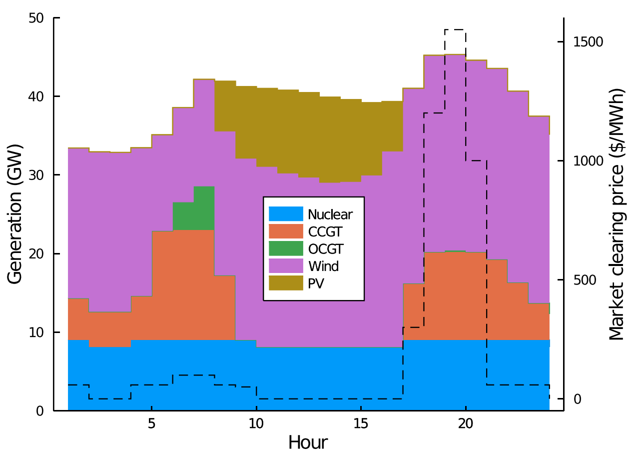

Fig. 8. Production by technology type and FCP market-clearing price for Solution 50.

Table 3
Comparison of market-clearing prices.

FCP ($/MWh)		
Hour	Solution 1	Solution 50
17	58.5	300.0
18	99.4	1200.0
19	250.0	1550.0
20	150.0	1000.0

producer surplus. The effect of the decreased production in Solution 50 is overpowered by the large increase in price, leading to an overall increase in producer surplus compared to Solution 1. Demand elasticity is an important driver of the magnitude of these results. In a long-run equilibrium setting with sufficiently price-responsive demand, demand sets the price fairly often, leading to many opportunities in which suboptimal commitment decisions might lead to lower cleared demand and thus higher market-clearing prices.

6. Conclusion

We compare several non-convex pricing models with the goal of determining to what extent it is worth pursuing smaller MIP gaps in electricity market-clearing algorithms from a social welfare perspective and how this conclusion varies by pricing model. Using a long-run adapted resource mix with price-responsive demand, we find a non-monotonic but nevertheless significant trend in changes in consumer and producer surplus with optimality gap of the UC dispatch

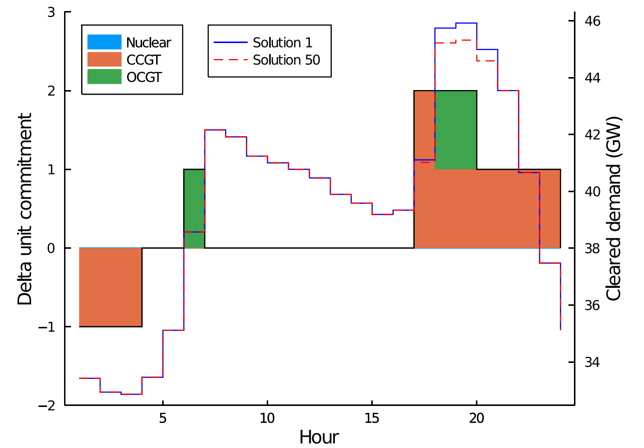

Fig. 9. Difference between unit commitment and cleared demand in Solution 1 and Solution 50.

decision for some pricing models. For pricing models dependent on the primal solution, such as locational marginal pricing, consumer surplus generally increases with increasing optimality and producer surplus correspondingly decreases. These results are influenced highly by scarcity rent from price-setting elastic demand. Convex hull pricing does not depend on the primal solution, yielding a high consumer surplus across near-optimal solutions. We also find that short-run cost recovery can be achieved over longer time horizons without any make-whole payments in a long-run adapted system. This finding raises the question of to what extent make-whole payments in real-world markets are necessary for long-run cost recovery or if they may distort the important economic signal to exit in the long-run that negative profits provide.

In the long-run, excess producer profits or losses associated with higher MIP gaps in some non-convex pricing models may distort the optimal resource mix away from the central planner mix. Future work could examine if the relationships between surplus and optimality gap hold for each pricing model in systems that are long-run adapted to each pricing model. There is also a trade-off between a desire to show a realistic test case with a large range of generator types and characteristics with the desire to perform an analysis in a long-run adapted equilibrium. Additional analysis is needed to examine if these results hold for a more diverse set of resources and over rolling horizons and multi-settlement markets. Further work may also employ a diversity criterion to ensure there is a good representation of the variety of near-optimal solutions that are possible. Finally, the impact of different demand elasticities on results should be explored. As energy and capacity market designs become more scrutinized with growing shares of VRE, there is a greater need to understand the role of near-optimal solutions and non-convex costs in electricity markets. This work provides some evidence that, particularly with the inclusion of price-responsive demand, small improvements in optimality may have significant positive effects on overall consumer and producer surplus.

CRediT authorship contribution statement

Conleigh Byers: Conceptualization, Data curation, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Gabriela Hug:** Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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